Week 4 – OSPF Single Area Lab Report

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Topology: R1 - SW1 - SW2 with VLAN 10,20,30 + R-EDGE LAN 172.16.1.0/24 via OSPF Area 0
Link: R1-R-EDGE 10.0.0.0/30 (R1 .1, R-EDGE .2)

Step 1: Remove Week 3 Static Routes

Before enabling OSPF, all static routes were removed to avoid conflicts.
On R-EDGE: no ip route 192.168.10.0 255.255.255.0 10.0.0.1 etc.
On R1: no ip route 172.16.1.0 255.255.255.0 10.0.0.2 and no ip route 0.0.0.0 0.0.0.0 10.0.0.2
Verification: show ip route no longer shows S routes.

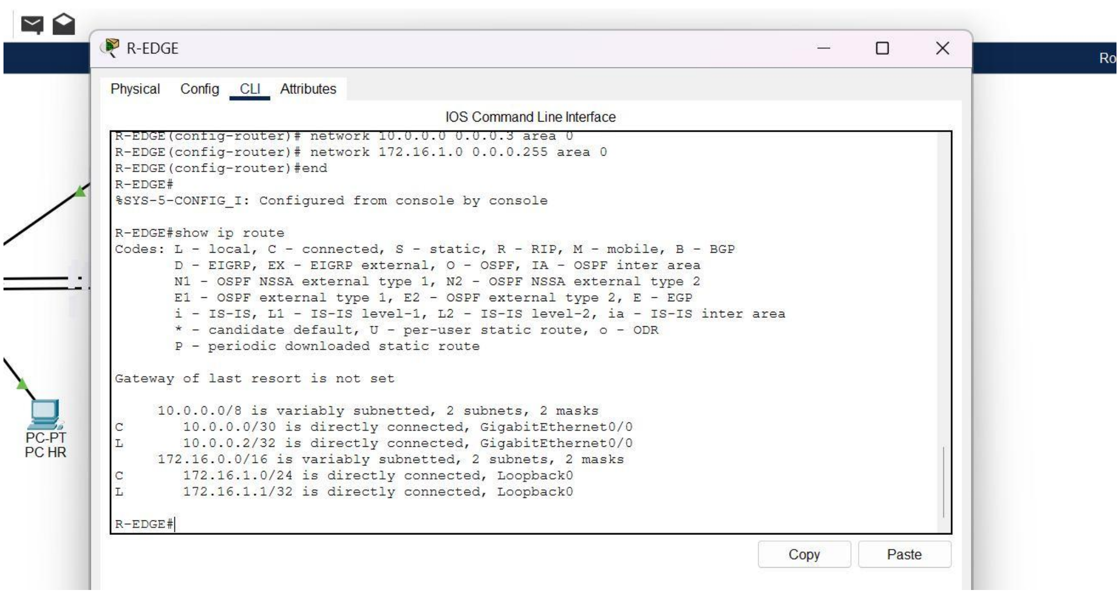
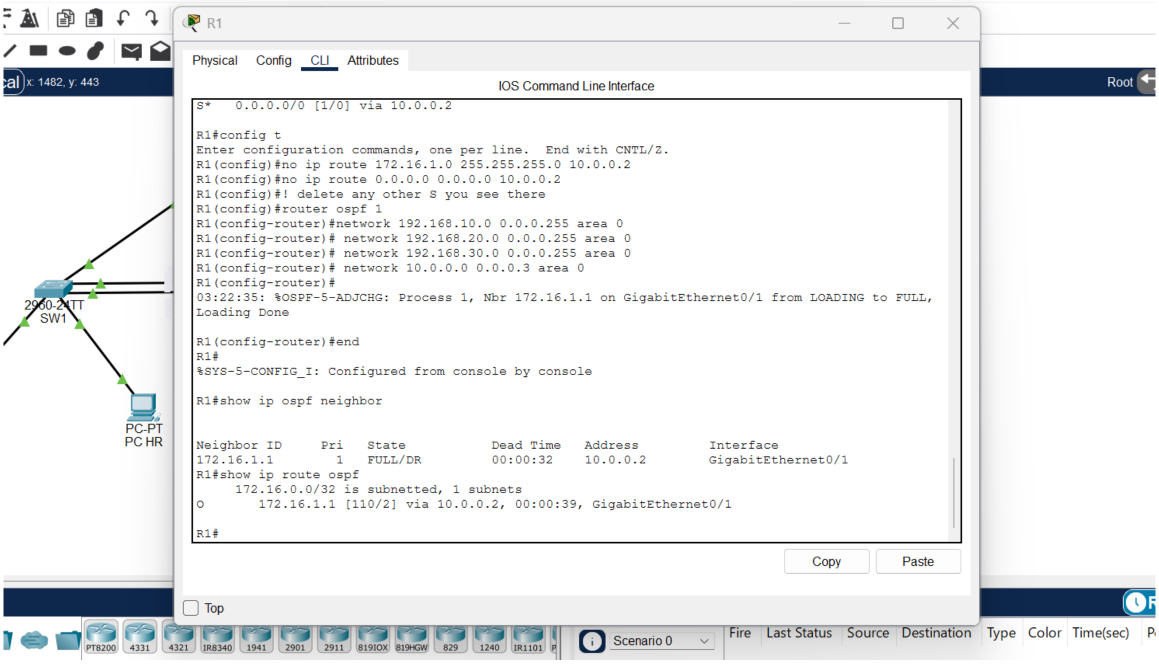


Figure 1: R-EDGE after removing S routes – only C and L routes remain. OSPF networks configured: 10.0.0.0 0.0.0.3 area 0 and 172.16.1.0 0.0.0.255 area 0

Step 2: Enable OSPF Area 0

32. On R1:
router ospf 1
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

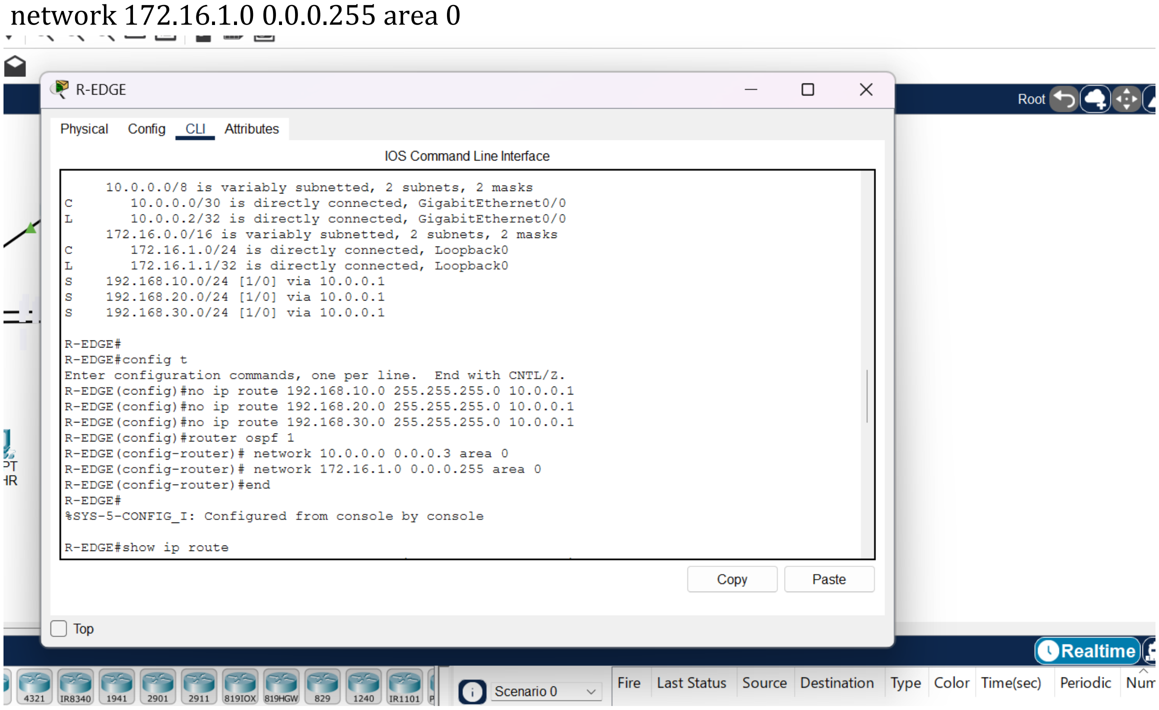


33. On R-EDGE:

router ospf 1

network 10.0.0.0 0.0.0.3 area 0

network 172.16.1.0 0.0.0.255 area 0



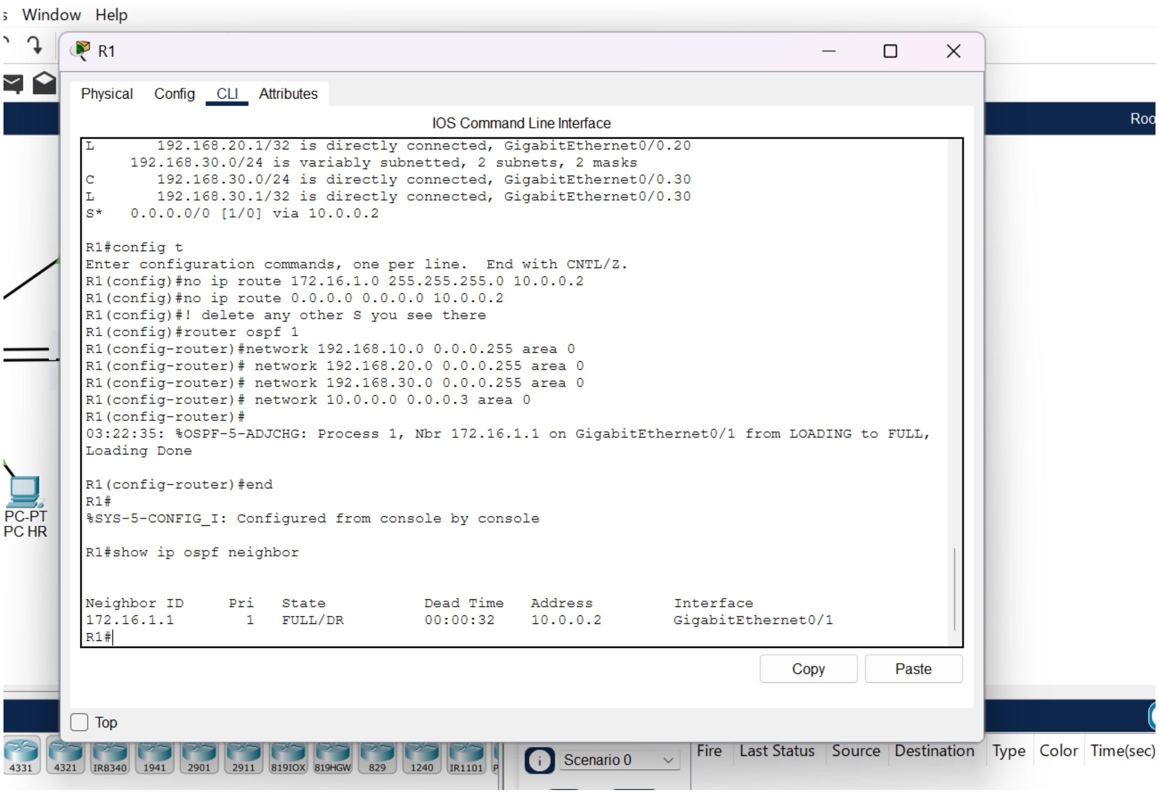
Wildcard calculation: /30 = 0.0.0.3, /24 = 0.0.0.255

Step 3: Verification

34. show ip ospf neighbor

Expected output on R1:
Neighbor ID Pri State   Dead Time Address   Interface
172.16.1.1   0 FULL/ -   00:00:33 10.0.0.2 GigabitEthernet0/0

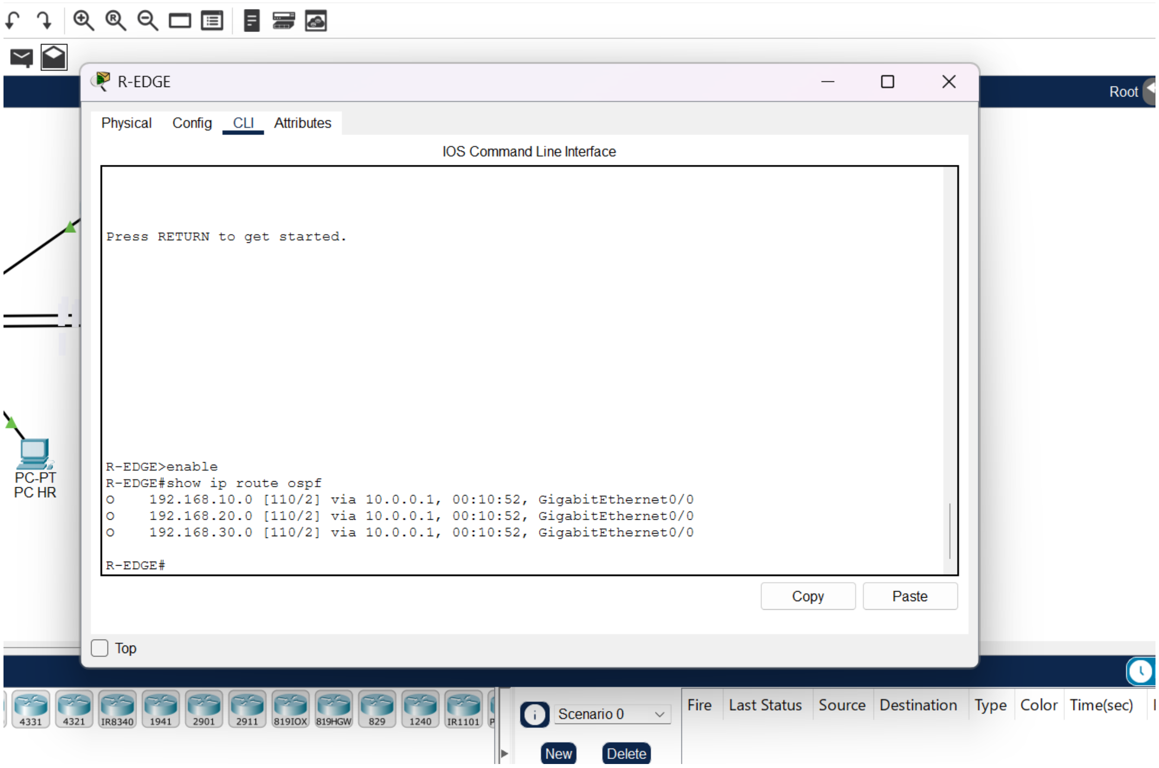
If State is FULL, adjacency is successful. If empty, check network statements and that no static default blocks OSPF.



35. show ip route ospf

Expected output on R1:
O 172.16.1.0/24 [110/2] via 10.0.0.2, GigabitEthernet0/0

Expected output on R-EDGE (previously you had S routes, now should be O):
O 192.168.10.0/24 [110/2] via 10.0.0.1
O 192.168.20.0/24 [110/2] via 10.0.0.1
O 192.168.30.0/24 [110/2] via 10.0.0.1



36. End-to-End Ping – VLAN PC to R-EDGE LAN

Test: PC SALES (VLAN10 192.168.10.x) -> 172.16.1.1 (R-EDGE Loopback0)

Result: Success – 0% loss, TTL 254, proves inter-VLAN routing + OSPF works.

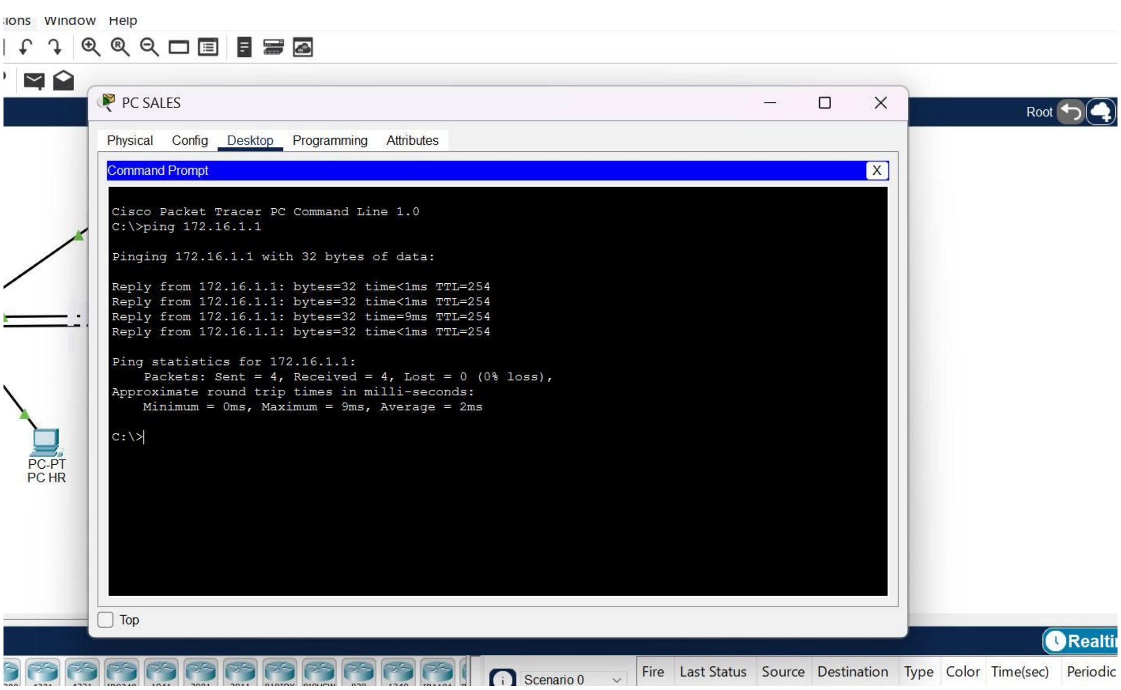


Figure 2: PC SALES ping 172.16.1.1 – Reply bytes=32 time<1ms TTL=254, Packets Sent 4 Received 4 Lost 0 (0% loss)

Reflection Q37

Compare troubleshooting a static-route failure (Week 3) to an OSPF neighbor failure (this week)
Static-route failure was harder to diagnose.
In Week 3, a static route failure is silent. show ip route simply does not display the S route, or it displays it but points to an unreachable next-hop. There is no log, no neighbor state. You must manually verify: 1) ip route syntax, 2) next-hop IP reachable (ping next-hop), 3) exit interface up/up, 4) every router along the path has a return route. A typo (e.g., 10.0.0.12 instead of 10.0.0.2) gives no error message.